

PORTO SANTO AP, Portugal

WMO: 085240

Lat: 33.0714N

Lon: 16.3486W

Elev: 272

StdP: 14.55

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	53.5	55.1	41.1	38.3	58.3	42.9	41.1	58.8	28.8	59.8	26.2	59.7	7.5	340	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	79.0	70.4	77.4	69.4	76.8	69.0	72.6	76.3	71.5	75.5	70.5	74.7	11.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.5	117.6	74.7	70.0	111.5	73.8	68.9	107.2	73.3	36.3	76.0	35.3	76.0	34.5	74.5	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.0	21.1	19.0	DB	49.9	83.9	1.8	4.0	48.6	86.8	47.5	89.1	46.5	91.3	45.2	94.2	
			WB	46.0	73.8	1.4	1.7	45.0	75.1	44.1	76.1	43.3	77.0	42.3	78.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.3	60.9	60.4	61.3	62.5	65.0	68.5	71.3	73.3	72.6	70.2	66.0	63.1	(6)
	DBStd	5.05	2.42	2.22	2.42	2.09	2.19	2.01	1.87	1.55	1.86	2.22	2.78	2.40	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	569	127	128	118	79	27	1	0	0	0	1	20	68	(9)
	CDD50	5949	339	292	349	375	464	555	661	721	679	627	481	406	(10)
	CDD65	1043	1	1	3	4	26	105	196	256	229	162	51	9	(11)
	CDH74	1336	0	0	0	3	7	36	211	529	425	119	6	0	(12)
	CDH80	38	0	0	0	0	0	1	10	12	13	2	0	0	(13)
(14) Wind	WSAvg	10.5	10.2	10.6	11.3	10.9	10.9	10.6	11.0	10.2	9.1	9.6	10.5	10.5	(14)
(15) Precipitation	PrecAvg	14.8	2.1	1.7	1.7	0.9	0.5	0.3	0.1	0.2	0.7	2.0	2.1	2.5	(15)
	PrecMax	23.5	7.1	4.4	5.0	2.3	2.7	1.3	0.5	0.6	2.8	6.2	7.0	8.0	(16)
	PrecMin	8.1	0.3	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.4	(17)
	PrecStd	3.8	1.9	1.1	1.3	0.6	0.5	0.3	0.1	0.2	0.6	1.7	1.5	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	68.3	70.5	72.0	74.3	77.1	80.4	81.0	81.0	78.6	75.0	70.6	(19)
		MCWB	63.1	61.9	61.4	62.5	65.5	69.0	69.1	71.1	70.6	70.0	68.1	64.6	(20)
	2%	DB	66.8	66.6	68.1	69.4	71.7	75.0	77.3	79.0	78.8	76.7	73.2	69.4	(21)
		MCWB	62.0	60.9	61.6	62.2	64.3	67.5	69.2	70.7	70.7	69.2	67.4	64.0	(22)
	5%	DB	66.1	65.7	66.5	67.9	70.1	73.5	75.9	77.4	77.3	75.3	71.6	68.1	(23)
		MCWB	61.4	60.3	60.6	61.2	63.0	66.1	68.0	69.5	69.5	68.1	65.9	62.9	(24)
	10%	DB	64.7	64.4	65.5	66.4	69.4	72.1	75.0	76.9	76.6	73.8	70.1	66.8	(25)
		MCWB	60.1	59.2	59.8	59.9	62.4	65.2	67.3	69.0	68.9	67.1	64.7	61.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.4	64.1	65.2	64.8	67.3	70.4	72.4	74.2	74.8	72.5	70.5	67.0	(27)
		MCDB	67.0	66.7	67.4	68.8	71.9	74.7	76.6	77.7	76.9	75.3	72.8	68.8	(28)
	2%	WB	63.8	62.6	63.5	63.4	65.6	68.9	70.9	72.5	72.8	71.0	68.9	65.5	(29)
		MCDB	65.8	65.6	66.4	67.5	69.7	73.1	75.0	76.7	76.1	74.3	71.6	67.9	(30)
	5%	WB	62.5	61.5	62.1	62.2	64.4	67.6	69.8	71.3	71.4	70.0	67.4	64.3	(31)
		MCDB	65.1	64.5	65.2	66.3	68.7	71.9	73.9	75.9	75.4	73.4	70.3	67.0	(32)
	10%	WB	61.3	60.1	60.9	61.1	63.4	66.7	68.7	70.2	70.2	69.0	66.0	63.2	(33)
		MCDB	64.1	63.4	64.3	65.3	67.8	71.0	73.4	74.9	74.5	72.7	69.3	66.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.3	7.7	8.3	8.1	7.9	7.9	7.9	8.2	7.9	7.1	6.8	(35)
		MCDBR	7.7	8.6	8.8	9.6	9.0	8.7	8.7	8.6	9.0	8.8	7.9	7.6	(36)
	5% WB	MCWBR	5.9	6.3	6.0	6.0	5.4	5.0	5.1	4.9	5.2	5.3	5.6	5.7	(37)
		MCWBR	7.0	8.1	8.3	9.1	8.8	8.3	8.8	8.5	8.8	8.0	7.2	6.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.337	0.358	0.384	0.375	0.379	0.379	0.387	0.390	0.387	0.372	0.362	0.343	(40)
		taud	2.562	2.487	2.406	2.448	2.461	2.483	2.466	2.476	2.501	2.534	2.523	2.555	(41)
	Ebn at Noon		279	284	283	290	288	286	284	282	279	275	268	270	(42)
		Edh at Noon	26	31	36	36	36	35	35	34	32	29	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		871	1140	1516	1887	2042	2173	2145	2027	1726	1323	931	789	(44)
	RadStd		42	85	83	72	148	127	121	100	60	55	57	42	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page